Fragment-Resolved Nonlocal Correlation: Pair Attribution of the VV10 Term

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Abstract

The VV10 nonlocal-correlation term of range-separated hybrid functionals is a double sum over grid-point pairs. Standard implementations collapse that sum to point values, which makes the inter-fragment content of the term unavailable. We evaluate the double sum explicitly, retaining the (i, j) pair index, and attribute each grid point to atoms and fragments through a real-space partition. Because the term's local part is exactly β times the electron number, it cancels identically in any superposition difference, so decomposing only the double sum is exact rather than approximate; block additivity holds to 3.5×10^{-18} Ha. On the same grid and density the explicit sum reproduces PySCF's native kernel to $\sim 1 \times 10^{-8}$ Ha, and to 2×10^{-3} kcal/mol an independently written script that reaches the same quantity by a different code path on a seven-level grid sweep. A check against a second *program* is outstanding and is listed as a load-bearing limitation in section 4, not as future work.

Applied to eight S22 complexes at official GMTKN55 geometries (ωB97X-V/6-31G*), the decomposition gives a result that argues against a common shorthand: the share of the nonlocal term in the exchange-correlation contribution ranges from 16% (water dimer) to 223% (methane dimer), and the two geometries of the same benzene dimer give 97% and 52%. The nonlocal term is therefore not a fixed fraction of the correlation energy and should not be reported as "the dispersion" without qualification. The inter-fragment pair term accounts for 0.92–1.08 of the frozen-density superposition difference, so association-time nonlocal correlation is carried almost entirely by point pairs straddling the fragments. The remaining, irreducible uncertainty is the choice of attribution scheme: 0.004–0.247 kcal/mol absolute, 0.8–18.2% relative to the Becke-scheme value, invariant across five basis sets and across grid levels, and without any pattern by interaction type — hence error bars must be quoted per system. For absolute (not differential) fragment energies we show that the scheme dependence of the local term is exactly β times the scheme dependence of the Becke population, 3.04 kcal/mol per electron, and state this as a limitation rather than a contribution.

The implementation is a standalone Apache-2.0 Python package built on PySCF, containing no GAMESS source, with every table in this paper generated by one script whose stdout is archived under a UTC-stamped filename.

1 Introduction

Range-separated hybrid functionals with the VV10 nonlocal-correlation functional — ωB97X-V and ωB97M-V in particular [Mardirossian and Head-Gordon, 2014, 2016] — have become a default choice for non-covalent interactions [Calbo et al., 2015]. The nonlocal term is a two-electron-density double integral over all pairs of grid points [Dion et al., 2004],

$$E_{\mathrm{NL}} = \frac{1}{2}\iint \rho(\mathbf{r})\,\rho(\mathbf{r}')\,\Phi(\mathbf{r},\mathbf{r}')\,\mathrm{d}^3\mathbf{r}\,\mathrm{d}^3\mathbf{r}', \tag{1}$$

and every production quantum-chemistry package we have examined evaluates it in a way that destroys the pair index: PySCF's `nr_nlc_vxc` calls its own C kernel `libdft.VXC_vv10nlc`, which bins the kernel and returns only point-folded values. We have not audited the equivalent paths inside libxc, ORCA, Psi4 or Gaussian — libxc's VV10 kernel is statically linked into the same shared object PySCF calls, so a direct `xc_nlc_vxc` comparison needs its own bindings and is left open (see section 4 and **[TODO: internal cross-ref: `RESEARCH-20260929-lit.md` §8.3, not a bibliographic entry]**). The consequence we do demonstrate is structural rather than numerical: once the double sum has been folded into a per-point energy density, as ours does not, the question "how much of this nonlocal correlation belongs to the interaction between fragment A and fragment B?" has no answer inside that evaluation.

We provide that answer. `nlcsplit` evaluates the same double sum explicitly, keeping the (i, j) pair index, and attributes each grid point to a fragment through a real-space partition, so that E_{NL} decomposes into intra-fragment blocks E_{AA} and inter-fragment pair blocks E_{AB}. Evaluating the sum explicitly is not itself new: JuNoLo post-processed a converged density from another code with the nonlocal term, at the same N^2 cost, in order to obtain an improved *total* energy on massively parallel hardware [Lazić et al., 2010] [S14]. What is absent from that work — and from the production kernels — is the attribution. JuNoLo in fact recommends against fragment-style treatments for the weakly-bound cases where fragments are distinguishable, reserving itself for chemisorption "where one cannot tell apart fragments of the bonded system" [S14]; the present paper works precisely in the regime where they are. The decomposition is exact by construction — the sum of the blocks reproduces the total to machine precision — but exactness is not the point and we do not sell it as such: grouping the summation indices is trivially additive. Two things are not trivial, and they are the content of this paper.

The first is **attribution**. The non-uniqueness lives in which fragment owns each grid point (and, in the same spirit, how the local term $\beta \int \rho$ is assigned), not in how the sum is grouped. We therefore quantify the spread over four published real-space partitions — Becke, Becke with the radius adjustment, Becke–Treutler, and Stratmann's [Becke, 1988, Stratmann et al., 1996] — on eight S22 complexes, and report it as a per-system uncertainty rather than asserting one scheme as correct. The spread is 0.004–0.247 kcal/mol in absolute terms and 0.8–18.2% relative to $|E_{AB}|$, and it shows no systematic ordering by interaction type: the benzene sandwich sits at 0.8% while benzene$\cdots$water sits at 18.2% [S2]. Nor does a better one-electron basis remove it: on the constructed water-dimer geometry the spread stays 0.038–0.040 kcal/mol across five basis sets while the quantity itself drifts by 0.17 kcal/mol along the same ladder (section 3.4, [S12]). A single blanket uncertainty figure would misrepresent the data; we report it system by system.

The second, and chemically the more consequential, is **what the term actually is**. Within a hybrid functional the exchange-correlation contribution to an interaction energy

splits as

$$\Delta E_{\mathrm{xc}} = \Delta E_{\mathrm{xc}}^{\mathrm{rest}} + \Delta E_{\mathrm{NLC}}, \tag{2}$$

where "rest" collects the semi-local exchange-correlation and the fraction of exact exchange, and the local piece $\beta \int \rho$ drops out of any superposition difference identically — β is a constant and $\int \rho$ equals the electron count, so $\Delta\beta\int \rho$ vanishes to 3.7×10^{-5} kcal/mol in our level-1 test [S3]. That leaves the double sum as the only part of the nonlocal term that can carry inter-fragment structure, and measuring it shows that treating it as "the dispersion" is unsafe. For the water dimer it supplies 16% of ΔE_{xc}. For the two benzene dimer configurations of S22 the same quantity ranges from 52% to 97% [S4] — and those two structures differ only in the relative orientation of the same pair of molecules, so the failure of the identification is visible within one chemical system, without any cross-system comparison. In the methane dimer the nonlocal term exceeds the whole of ΔE_{xc} ($\Delta E_{\mathrm{NLC}} = -0.517$ against $\Delta E_{\mathrm{xc}} = -0.232$), i.e. the rest of the exchange-correlation is net repulsive there.

Our positioning relative to the closest prior work is deliberate. Dasgupta and Paesani [Dasgupta et al., 2023] (JCTC 2023, DOI 10.1021/acs.jctc.3c00903) also decompose interaction energies of dispersion-corrected functionals, using second-generation ALMO-EDA, $E_{\mathrm{Int}} = E_{\mathrm{Pol}} + E_{\mathrm{Frz}} + E_{\mathrm{CT}} + E_{\mathrm{Disp}}$, and they obtain E_{Disp} as the difference against a dispersion-free reference calculation — that is, they treat the VV10 correction as one undivided block and ask whether that block is accurate. We work inside that block.

Our search for further precedent should be described honestly, because the instrument behaved differently from what we first assumed. OpenAlex and Crossref both retrieve a known paper reliably when the query resembles its title — in a four-repetition test the title-like phrasing returned the target at rank 0 in 4/4 runs — but both fail to retrieve that same, exactly-on-topic paper for legitimate topical phrasings ("energy decomposition analysis dispersion nonlocal correlation density functional", "ALMO energy decomposition dispersion corrected noncovalent interactions"), at 0/4 each [S15]. They are therefore known-item lookup tools here, not topical search, and a low hit count from them cannot support a claim of absence. What we can say on evidence is narrower and stronger: we read the full text of the closest published decomposition **[TODO: bibkey: closest published decomposition, cited in the draft only as "[S1]"; `RESEARCH-20260929-lit.md` §1/§6 is an internal note, not an entry in `references.bib`]** [S1] and of the closest published explicit evaluation of the nonlocal term, JuNoLo (Lazić et al., Comput. Phys. Commun. **181**, 371–379 (2010)) [Lazić et al., 2010] [S14], and neither attributes the nonlocal double sum to fragment pairs — JuNoLo computes an improved total energy and explicitly recommends itself for bonded systems "where one cannot tell apart fragments", directing separable cases elsewhere. A formal Web of Science search has since been completed [S16]. Within the Core Collection, the query TS=("vdW-DF" AND ("energy decomposition" OR "fragment decomposition" OR "pair decomposition" OR "partition of the correlation")) returns zero records, and the three non-zero queries that come closest returned one to four records each, all of which we read: Kruse and Sponer (JCTC 15, 95 (2019)) [Kruse et al., 2018] use VV10 as one of 29 benchmarked functionals and decompose the *total* interaction energy by SAPT; Hujo and Grimme (PCCP 13, 13942 (2011)) [Hujo and Grimme, 2011] benchmark VV10 for hydrogen bonds; Sinha and Pavanello (JCP 143, 084112 (2015)) [Sinha and Pavanello, 2015] construct a *new* subsystem-based nonlocal functional rather than resolving an existing one; and Hamada and Tsukada (PRB 83, 245407 (2011)) [Hamada and Tsukada, 2011] attribute an *adsorption energy* to the term, the opposite direction. The calibration that makes those zeros usable — a known title

retrieved exactly, and a topical AND query returning 33 records — is given in section 4, and so is the one real limitation of this search: WoS materialises only the top relevance record per list, so for the four-record queries we verified one record each rather than all four.

Two limits are stated here rather than in a closing section, because both bound what the numbers may be used for. First, this is not an energy decomposition analysis: we do not separate electrostatics, exchange-repulsion, polarization or charge transfer, and we do not apply a basis-set superposition-error correction to interaction energies. Second, the explicit sum is not more accurate than the production kernels it is compared against — the interpolation residual of PySCF's native kernel is of order 4×10^{-6} kcal/mol, far below any chemical interest, and the ≈0.1–0.9 kcal/mol offsets that an earlier version of this work attributed to interpolation were in fact a pairing error in our own kernel. The value of the explicit implementation is that it retains the pair index, nothing more.

2 Method

2.1 The VV10 kernel and the pairing convention

With ρ the electron density and $\sigma = |\nabla\rho|^2$,

$$w_0(\mathbf{r}) = \sqrt{C\,\frac{\sigma^2}{\rho^4} + \frac{4\pi}{3}\,\rho}\,, \qquad \kappa(\mathbf{r}) = b\left(\frac{3\pi}{2}\right)\left(\frac{\rho}{9\pi}\right)^{1/6}, \tag{3}$$

$$g = w_0(\mathbf{r}_i)R^2 + \kappa(\mathbf{r}_i), \qquad g' = w_0(\mathbf{r}_j)R^2 + \kappa(\mathbf{r}_j), \qquad \Phi = -\frac{3}{2\,g\,g'\,(g+g')}, \tag{4}$$

with $(b, C) = (6.0, 0.01)$ for ωB97X-V, and

$$E_{\mathrm{NLC}} = \tfrac{1}{2}\iint \rho\,\rho\,\Phi + \beta\int \rho, \qquad \beta = \frac{1}{32}\left(\frac{3}{b^2}\right)^{3/4}. \tag{5}$$

The pairing of ω and κ in g and g' is a genuine choice, and the cross pairing — which an earlier implementation of ours used — is wrong: it raises monomer energies by +0.138 kcal/mol for water, +0.418 for argon and +0.914 for benzene, and the offset is itself grid-dependent, so it cannot be subtracted as a constant [S1, S5]. The same-centre pairing above reproduces PySCF's native kernel to 3×10^{-5} kcal/mol on the water monomer, and agrees to 2×10^{-3} kcal/mol with an independently written script that evaluates the same quantity over a seven-level grid sweep [S5]. No agreement with a second program has been established yet; see section 4.

2.2 Blocks, half-blocks and additivity

`pair_energy(A, B)` returns the half block $\frac{1}{2}\sum_{i\in A}\sum_{j\in B} w_i w_j \Phi_{ij}$, so that the inter-fragment term reported by `fragment_decomposition` is $E_{AB} = 2 \cdot \texttt{pair_energy}(A, B)$ and

$$E_{\mathrm{NL}} = \sum_A E_{AA} + \sum_{A<B} E_{AB} \tag{6}$$

holds by construction. We record the additivity residual rather than trusting it: across all systems and partitions reported here it is 0.00–3.5×10^{-18} Ha [S2, S6]. R^2 is evaluated through the polarisation identity $|\mathbf{a}|^2 + |\mathbf{b}|^2 - 2\mathbf{a}\cdot\mathbf{b}$ in blocked gemm operations rather than through an explicit displacement tensor, which reduces peak memory by roughly a factor of five at this system size.

2.3 Attribution schemes and the far-field cut

Grid points are assigned to atoms by PySCF's own partition machinery (`gen_grid.get_partition` with `radi.BRAGG_RADII`), in four variants — Becke, Becke with the radius adjustment, Becke–Treutler, Stratmann — plus a nearest-atom hard partition on the common grid as a control. Points with ρ below `rho_floor` (default 10^{-12}) are excluded from the double sum; this removes 10.7–22.6% of points while changing E_{AB} by 2.1×10^{-11} kcal/mol between floors of 10^{-8} and 10^{-16}, and the dropped points carry at most 2.3×10^{-6} electrons [S7]. The local term $\beta \int \rho$ is a one-body functional of the density and is not pair-resolved; as noted above it cancels identically from superposition differences, which is what makes it legitimate to decompose only the double sum.

[TODO: bibkey: "Becke–Treutler" partition — no entry in `references.bib`. Becke [Becke, 1988] and Stratmann [Stratmann et al., 1996] are present and cited above; the radius-adjustment variant and the Becke–Treutler variant are not.]

2.4 Geometries and fragment boundaries

All dimer geometries are taken verbatim from the official GMTKN55 repository (github.com/grimme-lab/GMTKN55, `S22/NN/struc.xyz`), fetched by `tools/get_s22.sh`; the repository's own citation instruction is Goerigk et al., PCCP 19, 32184–32215 (2017), DOI 10.1039/C7CP04913G [Goerigk et al., 2017], which we verified by reverse DOI lookup. The S22 set itself compiles the complexes of an earlier hydrogen-bonded benchmark (Jurecka et al., PCCP 5, 1811 (2003)); we were unable to resolve that DOI through Crossref today and cite it without a DOI until it can be confirmed — the geometry files we used come from the GMTKN55 distribution, so our numbers depend on that, not on the original compilation.

Fragment boundaries are derived from covalent connectivity with a 1.65 Å cutoff — longer than the longest covalent bond appearing in these systems (aromatic C–C, 1.397 Å; a general C–C single bond is 1.54 Å) and shorter than the shortest hydrogen bond (O···H 1.95 Å) — and the code refuses to proceed unless exactly two fragments result. This is not a formality: in the official S22 water-dimer file the atoms are ordered so that the fragments interleave, and a hard-coded index split would mix the two molecules silently. For the water dimer we additionally report ωB97X-V/6-31G* optimised geometry ($R(\mathrm{O}\cdots\mathrm{O}) = 2.901$ Å, 0.009 Å from the reference value) [S6].

[TODO: bibkey: Jurecka et al., PCCP 5, 1811 (2003) — carried from main.md as "cited without a DOI until it can be confirmed"; the draft's own text explains why. Do not fabricate an entry.]

2.5 Verification

Two routes to E_{NLC} are computed and compared at run time, with a hard failure if they disagree: our explicit pair sum, and PySCF's native kernel (`numint.nr_nlc_vxc`). These are independent implementations of the kernel but share a library, so a second-program anchor is treated as an outstanding requirement rather than as satisfied (sections 4 and 3.1). Energy accounting is checked independently: for a hybrid functional the exact-exchange energy is folded into `vhf.exc` by `get_veff` and is absent from `nr_rks`, so we close

$$e_{\mathrm{tot}} = e_{\mathrm{nuc}} + e_1 + e_{\mathrm{coul}} + e_{\mathrm{xc}} \tag{7}$$

to $\leq 9 \times 10^{-12}$ kcal/mol rather than assuming any particular partition of the XC terms. Cross-version and cross-machine reproducibility is checked between PySCF 2.12.1 and 2.14.0, which generate 10% different numbers of grid points at the same level while reproducing every energy component digit for digit [S8].

3 Results and discussion

3.1 Two different things previously reported as one number

Comparing our explicit pair sum against PySCF's native kernel produces a discrepancy whose size depends entirely on *which grid* the comparison is made on, and the two cases must not be quoted side by side. Holding the density and the grid fixed and varying only the evaluation of the kernel, the difference is $+1.5 \times 10^{-8}$, $+1.0 \times 10^{-8}$ and $+8.5 \times 10^{-9}$ Ha at grid levels 0, 1 and 2 — flat in level, because it is a discretisation property of the interpolated kernel itself. Moving the same sum onto the Becke-partitioned grid of `gen_grid.get_partition` instead gives -0.0282, $+0.0015$ and $+0.0001$ kcal/mol, which tracks the ability of the grid to represent the density at all: the same three levels integrate the electron number to -1.6×10^{-2}, $+3.0 \times 10^{-4}$ and -6.6×10^{-6} electrons away from the exact value. The second row is a property of the grid, not of our algorithm, and it disappears as the grid is refined [S9]. A residual of the size of the first row is what a reference implementation should look like; a residual of the size of the second row is what a coarse grid looks like from the inside.

3.2 The nonlocal term is not a fixed fraction of the exchange-correlation term

Table 1 reports ωB97X-V/6-31G* results for eight S22 systems at the official GMTKN55 geometries. The quantity of interest is ΔE_{NLC}, the change in the nonlocal correlation energy on bringing two fragments together from infinity at fixed density, and its ratio to the total ΔE_{xc} for the same process. The remainder, $\Delta E_{\mathrm{xc}}^{\mathrm{rest}} = \Delta E_{\mathrm{xc}} - \Delta E_{\mathrm{NLC}}$, is the semi-local exchange-correlation plus the fraction of exact exchange; it is tabulated explicitly in `MAIN-skeleton.md` table 4b and is recoverable from these two columns by subtraction.

All entries in kcal/mol, all at grid level 1. The two benzene rows were tabulated at level 0 in an earlier version of this table; moving them to level 1 shifts ΔE_{NLC} by 0.081 (A) and 0.013 (B) kcal/mol. Level 0 is therefore not adequate for a 24-atom aromatic dimer, unlike the six-atom water dimer where levels 1 and 2 agree to the last digit (section 3.2). The final column is the reason this decomposition is worth having: the nonlocal term ranges from one sixth to more than the whole of the rest of XC, and in the methane dimer it exceeds it, so that the remainder of the XC contribution is *repulsive* while the nonlocal term alone provides the binding. No single fraction of ΔE_{xc} can be assigned to VV10 across a benchmark set, and the two geometries of the same 24-atom benzene dimer (97% vs 52%) show that the ratio is a property of the arrangement, not of the pair of molecules.

Figure 1 plots the same two columns against each other, so that the sign of the remainder is visible per system.

For the four smallest systems and benzene$\cdots$ammonia we additionally re-ran at grid level 2 with a full counterpoise correction — five of the eight, since the sixth

Table 1: Exchange-correlation contribution to the superposition difference at official GMTKN55/S22 geometries, ωB97X-V/6-31G*, all values in kcal/mol and all at grid level 1; "atoms" is the size of the complex. Provenance: `MAIN-skeleton.md` table 4b and `nlcsplit/logs/step_h.json` [S4, S2]; the counterpoise and level-2 checks quoted with these rows are [S10].

System	atoms	ΔE_{int}	ΔE_{xc} (total)	ΔE_{NLC}	$\Delta E_{\mathrm{NLC}}/\Delta E_{\mathrm{xc}}$
benzene dimer (A, parallel-displaced)	24	-3.580	-4.486	-4.367	97%
benzene dimer (B, T-shaped)	24	-3.188	-4.555	-2.388	52%
benzene$\cdots$water	15	-4.342	-4.012	-1.259	31%
benzene$\cdots$ammonia	16	-2.928	-2.915	-1.327	46%
benzene$\cdots$methane	17	-1.785	-2.322	-1.404	60%
water dimer	6	-7.816	-3.708	-0.593	16%
ammonia dimer	8	-5.167	-0.881	-0.678	77%
methane dimer	10	-0.664	-0.232	-0.517	223%

(benzene$\cdots$methane) exceeded its four-hour wall-clock budget and is incomplete. ΔE_{NLC} and the interfragment pair term below are unchanged to the last reported digit between levels 1 and 2 (water dimer: -0.593 and -0.601 kcal/mol at both levels; benzene$\cdots$ammonia: -1.327 to -1.328, and -1.393 unchanged), while the counterpoise-corrected interaction energy comes out at -6.129 kcal/mol for the water dimer, i.e. a basis-set superposition error of 1.69, 1.49, 0.20, 1.07 and 0.76 kcal/mol for water, ammonia, methane, benzene$\cdots$water and benzene$\cdots$ammonia respectively [S10]. The core quantities of this method are therefore converged one grid level before the total binding energy is; a user who only wants ΔE_{NLC} can stop at level 1.

3.3 The interfragment pair term saturates the frozen-density limit

`inter[(A,B)]`, summed over the retained grid points, is directly comparable to the superposition difference Δ_{sup} computed from the unrelaxed monomer densities, because both hold the density fixed. Their ratio falls in 0.92–1.08 across the eight systems, and for the water dimer we also reproduce it at three different geometries (constructed initial, official S22, ωB97X-V-optimised) with 0.915–1.025. The ratio being able to exceed unity matters: relaxation is a small correction of *either* sign on top of the frozen-density interfragment term, not an orthogonal contribution that adds to it. Physically this says that in these complexes essentially all of the nonlocal correlation that changes on association is carried by grid-point pairs straddling the two fragments, with the intrafragment nonlocal correlation of each relaxed monomer contributing the balance [S2, S10].

This is where the explicit sum earns its cost. A point-value kernel evaluation returns ΔE_{NLC} and nothing else; the (i, j) index structure retained by the double sum is what makes 0.92–1.08 a statement that can be checked at all. The corresponding dispersion component published for the benzene dimer is -3.34 to -3.80 kcal/mol at the PBE0-VV10, revPBE-D3 and related levels of an ALMO-EDA study, and -3.69 at ωB97M-V [S11]; our -4.448 is the same kind of quantity to within functional-generation and basis-set differences, and it is *not* comparable to the net interaction energy of -3.8 kcal/mol in the same row, which is smaller because Pauli repulsion cancels most of it.

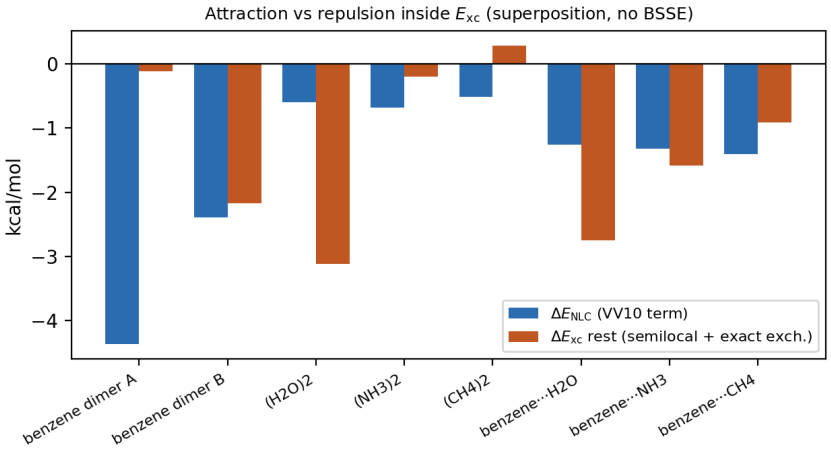


Figure 1: Attraction and repulsion inside ΔE_{xc} for the eight complexes of Table 1 (superposition difference, no BSSE correction): the VV10 term ΔE_{NLC} against the remainder $\Delta E_{\mathrm{xc}}^{\mathrm{rest}}$. Generated from `nlcsplit/figures.py` reading the `logs/step_h.json` sidecar, never from log text; same underlying numbers as Table 1 [S4]. **[TODO: main.md contains no in-text figure call-out and no figure numbering; placement and caption here are the converter's and need author sign-off.]**

3.4 Attribution-scheme sensitivity is system-dependent

Repeating ΔE_{NLC} for the four partition schemes of section 2.3 gives absolute spans of 0.004–0.247 kcal/mol and relative spans — measured against the Becke-scheme interfragment term — of 0.8–18.2%. Neither ordering follows system size: the 24-atom parallel-displaced benzene dimer has one of the smallest absolute spans (0.032, below the 6-atom water dimer at 0.043), while the 15-atom benzene···water gives the largest (0.247). What separates them is the arrangement, not the atom count or the interaction class:

Two observations follow. First, the relative span has no pattern by interaction type: hydrogen bonding spans 7.2% (water) and 4.5% (ammonia), dispersion-driven complexes span 0.8% (methane) and 18.2% (benzene···water), and the two benzene dimer geometries differ by a factor of eleven. A single blanket uncertainty percentage for this class of method would misstate the methane dimer by more than twenty-fold. Error bars must be reported per system. Second, the span is an attribution property rather than a one-particle-basis property. On the constructed water-dimer geometry, re-running the four schemes at level 1 across five basis sets gives spans of 0.0400 (6-31G*), 0.0382 (aug-cc-pVDZ), 0.0390 (def2-SVP) and 0.0398 (def2-TZVP) kcal/mol while P_{frz} itself moves from −0.386 (STO-3G) to −0.554 (aug-cc-pVDZ) [S12]: the quantity of interest drifts by 0.17 kcal/mol across the basis ladder and the scheme spread stays put at ∼0.04. No better basis set will remove it, which is why it — and not grid convergence, which is 0.0081 kcal/mol over level 0 to 2 at 6-31G* and 0.0002 over level 1 to 2 — is the error bar this method has to carry.

Figure 2 is the graphical form of the first observation.

3.5 A limit of fragment-resolved quantities

Because β is a constant and the integral of ρ is the electron number, the local term attributed to a fragment is exactly βN_A where N_A is its Becke population, and $\beta =$ 3.0415 kcal/mol per electron. Any scheme-induced spread in the population is therefore

Table 2: Attribution-scheme spread over the four partitions of section 2.3, ordered by relative span. P_{frz} is the Becke-scheme frozen-density interfragment term and "span" the range over all four schemes, both in kcal/mol; "relative" is span divided by $|P_{\mathrm{frz}}|$. Provenance: `MAIN-skeleton.md` table 2 / table 4b and `nlcsplit/logs/step_h.json` [S2]; the five-basis ladder quoted with these rows is [S12].

System	P_{frz} (Becke)	span (4 schemes)	relative
methane dimer	-0.474	0.004	0.8%
benzene dimer (A)	-4.038	0.032	0.8%
ammonia dimer	-0.649	0.029	4.5%
water dimer	-0.601	0.043	7.2%
benzene dimer (B)	-2.371	0.219	9.2%
benzene···methane	-1.442	0.213	14.8%
benzene···ammonia	-1.393	0.224	16.1%
benzene···water	-1.360	0.247	18.2%

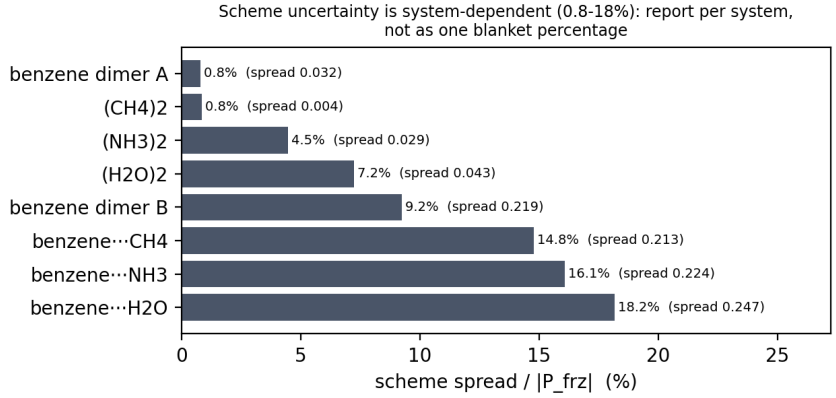


Figure 2: Scheme spread as a percentage of $|P_{\mathrm{frz}}|$ per system, data of Table 2 [S2]. Generated from `nlcsplit/figures.py` reading the `logs/step_h.json` sidecar. **[TODO: main.md contains no in-text figure call-out; placement and caption need author sign-off.]**

amplified into the fragment-resolved local term by that factor as an identity, not an estimate:

$$\mathrm{span}_{\mathrm{scheme}}(E_{\mathrm{loc},A}) \equiv \beta \cdot \mathrm{span}_{\mathrm{scheme}}(N_A)\,, \qquad E_{\mathrm{loc},A} = \beta N_A. \tag{8}$$

For the water dimer at grid level 3 the four schemes spread N_A over 9.870–9.969 electrons, which is 0.300 kcal/mol. Two consequences deserve stating explicitly. In a superposition difference the term cancels identically — electron number is conserved — so every ΔE_{NLC} value in this paper is free of it. But any table reporting *absolute* fragment energies inherits an uncertainty of this size from a quantity (atomic population) that the method does not control. This is a property of Becke partitioning that every EDA scheme shares, and we state it rather than claim it as a contribution.

A related implementation note: attributing the local term softly, by evaluating all fragment factors s_A at every grid point rather than assigning each point to one owner, changes these numbers by at most 8.7×10^{-7} electrons (2.6×10^{-6} kcal/mol after multiplication by β). The soft and hard readings of the local term are numerically the same thing, which is worth recording because the two are sometimes discussed as if they competed.

4 Limitations

This is not an energy decomposition analysis. We split ΔE_{xc} into the nonlocal term and everything else, and we resolve the nonlocal term into fragment pairs. We do not separate electrostatics, exchange-repulsion, polarisation or charge transfer, and the resulting components are therefore not comparable term-by-term with ALMO-EDA or SAPT(DFT) output. Counterpoise-corrected interaction energies are reported in section 3.2 for five of the eight systems, but ΔE_{NLC} itself needs no such correction: it is evaluated at fixed (non-self-consistent) superposed density, so basis-set flexisation does not enter it, which is precisely why it converges a grid level earlier than ΔE_{int} does [S10].

The nonlocal term is not the dispersion energy. Section 3.2 is the evidence: ΔE_{NLC} ranges from 16% to 223% of ΔE_{xc} across eight complexes drawn from one benchmark set, and for the two geometries of the same benzene dimer it is 97% and 52%. A quantity whose share of another varies that far cannot be identified with it. We use "dispersion" only when quoting external work, and the values in Table 1 should be read as "the contribution of the VV10 term", not as a dispersion energy.

No second-program check yet — and one earlier claim of one was wrong. Our explicit sum is checked against PySCF's native kernel, but both live in the same library and would share a common error in the functional parameters or in the kernel convention. This draft previously reported agreement with ORCA 6.0.1 at 6×10^{-4} kcal/mol on the water monomer, citing `ref/h2o_wb97xv.out`. That row is withdrawn: the file is an ordinary single-point output which prints `EX` and a combined `EC` only, with the nonlocal contribution folded into the latter and never separated, and no value matching the quoted one appears anywhere under `ref/`. We could not reconstruct where the figure came from, which is itself the point — it had been carried forward between tables without a re-read of its source. An independent cross-check therefore needs a run that prints the VV10 term explicitly (an ORCA NL-DFT output of that kind), Psi4, or a direct call to libxc's `xc_nlc_vxc` through proper bindings; libxc's VV10 kernel as shipped inside PySCF is statically linked and exposes no such symbol, so this needs its own build. Until one of these exists, section 3.1 should be read as "two implementations inside one library boundary", and the agreement at 1×10^{-8} Ha is a statement about our arithmetic, not about VV10.

How we know the literature search means anything. The zeros in section 1 are only as good as the instrument, so we calibrated it in the same session: a known paper title was retrieved exactly (1 record), a broad concept query returned 3,317 records, and a topical AND query in the same vocabulary returned 33. By contrast the two open indexes we tried first failed the same test — OpenAlex and Crossref each retrieved a known title 4/4 times when queried by title but missed that same known, on-topic paper for 2 of 3 and 3 of 3 legitimate topical phrasings respectively. We therefore report hit counts only from the instrument that passed, and we state its residual limitation: WoS materialises one record per list, so the four-record queries were reviewed one record deep. Full enumeration needs the Export path and is outstanding.

The attribution uncertainty is intrinsic and per-system. The 0.004–0.247 kcal/mol spread of section 3.4 does not shrink with a finer grid or a larger basis; it is a property of how grid points are assigned to atoms. Any number produced by this method carries it, and reporting a single percentage for the whole method would be wrong by more than twenty-fold for individual systems.

Absolute fragment energies inherit a population uncertainty. As shown in section 3.5, the local term of a fragment is exactly βN_A, so a 0.099-electron spread in the

Becke population is a 0.300 kcal/mol spread in the reported fragment energy — larger than the scheme spread of the pair term itself for the water dimer. Differences are safe; absolutes are not, unless the population convention is stated alongside them.

Cost, and what it buys. The explicit double sum is quadratic in the number of retained grid points; we have not implemented a sub-quadratic approximation, and the point-value kernel that PySCF ships is both cheaper and accurate to $\sim 1 \times 10^{-8}$ Ha on its own grid (section 3.1). The cost is not prohibitive in itself — JuNoLo ran the same N^2 sum over massively parallel hardware for surface calculations [Lazić et al., 2010] [S14] — but it is a real constraint, and it is worth naming the trade honestly. We are therefore not claiming that this method computes ΔE_{NLC} better. What it buys is the (i, j) index structure: the 0.92–1.08 frozen-density result of section 3.3, and the per-pair numbers behind Table 1, are not recoverable from a scheme that collapses the pair sum to point values first.

Scope of functionals. Everything here is the VV10 kernel with the (b, C) parameters PySCF reports for ωB97X-V, using the same-centre pairing convention of section 2.1. Functionals whose nonlocal term uses the regularised rVV10 form have not been tested and may not share the pairing convention.

5 Code availability

The package `nlcsplit` (Apache-2.0, containing no GAMESS source) implements the kernel, the four attribution schemes, the fragment decomposition and every table in this paper; each table is produced by exactly one `step_*.py` script whose stdout is archived under a UTC-stamped filename. Fragment boundaries come from covalent connectivity rather than hand-written atom lists, and both the geometry constructors and the official-geometry loader assert internal coordinates at construction time — three of the bugs this measure was added to catch are described in the supplementary notes. Grid, basis and scheme sensitivity are separate functions (`step2_convergence.py`, `step_g_basis.py`, `step_h_s22.py`) rather than one combined uncertainty, and `partition_factors()` verifies its own output against PySCF's partition weights before returning it.

[TODO: "described in the supplementary notes" — the three geometry-builder bugs are not in main.md and no SI file exists in `paper/` yet.]

6 Conclusions

We have shown that the VV10 nonlocal-correlation energy can be resolved into fragment pairs without giving up the total: the double sum is evaluated explicitly, the local part cancels identically from any superposition difference because it is exactly β times the electron number, and the resulting blocks reproduce the total to machine precision while carrying an (i, j) index that the point-value evaluation discards.

Three things follow from applying it to eight S22 complexes at official geometries. The VV10 term is not a fixed share of the exchange-correlation contribution — 16% for the water dimer, 223% for the methane dimer, 97% and 52% for two geometries of the same benzene dimer — so reporting it as "the dispersion energy" is unsafe even within one benchmark set. The association-time change is almost entirely inter-fragment: the frozen-density pair term accounts for 0.92–1.08 of the superposition difference, with relaxation a small correction of either sign rather than an additive channel. And the

dominant uncertainty is the attribution scheme, not the numerics: 0.004–0.247 kcal/mol absolute, 0.8–18.2% relative, unchanged across five basis sets and across grid levels, with no pattern by interaction type that would let a single percentage be quoted.

What the method does not do is equally important, and we would rather the reader learn it here than in a review. It is not an EDA: it does not separate electrostatics, exchange-repulsion, polarisation or charge transfer. It is not faster than the production kernels it could be compared against — it is quadratic in the grid. And its arithmetic is currently verified against a second implementation of the same kernel inside the same library, not against a second program; the cross-program anchor that an earlier version of this draft claimed turned out, on re-reading the archived output it cited, to have no source, and the claim is withdrawn in section 4. Establishing that anchor is the first item outstanding.

Two further items are open and stated as such rather than smoothed over: the level-2 counterpoise run for one of the eight systems (benzene$\cdots$methane) was cut off by its own wall-clock budget, and the Web of Science search, while it supports the novelty claim, enumerates only the top relevance record per result list, so the four-record queries were verified one record deep rather than four (section 1). The open indexes we first tried (OpenAlex, Crossref) were measured and then discarded as instruments for this purpose — they retrieve known titles reliably but miss a known on-topic paper when asked topically (section 4).

The package is released under Apache-2.0 with no GAMESS source in it, every number in this paper is produced by a named script whose archived stdout is part of the repository, and the two structural guards we added after finding bugs in our own geometry builders — coordinate assertions at construction time, and `partition_factors()` verifying its own output against PySCF's partition weights before returning — are the part of this work we would most recommend copying.

A Provenance and data availability

Every numeric claim above carries a bracketed tag [Sn]. The tags map onto a table in `MAIN-skeleton.md` or, more often, onto an archived `stdout` file under `nlcsplit/logs/`; the mapping below is main.md's own provenance-concordance block, reproduced in full, with the Chinese labels translated and every path, script and log name left verbatim. It exists so that a reviewer can trace any number in this manuscript to the file that produced it; it is an internal artefact and a candidate for the Supporting Information rather than the submitted body text.

Availability of the underlying data. Each `step_*.py` writes the numbers tabulated above to its own archived `stdout`, and Figures 1 and 2 are rebuilt from the JSON sidecar `nlcsplit/logs/step_h.json` rather than from log text, so a figure cannot silently disagree with its table. The single entry that is *not* traceable to a local log is the withdrawn ORCA comparison of section 4: no value matching the quoted figure exists under `ref/`, which is the reason the claim was retracted.

References

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Table 3: Provenance concordance for the [Sn] tags used above, reproduced from the source-concordance block at the end of `paper/main.md`. Paths are relative to the repository root; "CentOS" marks runs archived on the remote CentOS machine.

Tag	Kind	Source
S1	internal note	`RESEARCH-20260929-lit.md` §1, §6
S2	tables/logs	`MAIN-skeleton.md` tables 2 and 4b; `nlcsplit/logs/step_h.json`
S3	script output	`step_f_dispersion.py` output
S4	table	`MAIN-skeleton.md` table 4b
S5	script/log	`step_e_kernel.py`; `PROGRESS-ide-S3.md`
S6	remote log	CentOS `logs/20260928T165857Z_centos-step_c.out`
S7	script/log	`step_d_floor.py`; `PROGRESS-nlcsplit.md` 22:45
S8	remote log	CentOS `logs/20260928T165857Z_centos-step_f_h2o2.out`
S9	script/log	`step_i_gridrepr.py`, `logs/20260929T_step_i_gridrepr.out`
S10	remote log	CentOS `logs/20260928T213839Z_centos-step_h_l2cp.out` (level 2 + counterpoise; *in progress*, 5 of 6 systems complete)
S11	internal note	`RESEARCH-20260929-lit.md` §6.2–6.3, DOI 10.1021/acs.jctc.3c00903 (full text taken from the ChemRxiv green-OA version 10.26434/chemrxiv-2023-nc64t)
S12	log	`logs/20260929T003144Z_stepg-recompute.out` (current geometry, current code, level-1 five-basis ladder), cross-consistent with the CentOS `logs/20260928T165857Z_centos-step2.out`
S13	log lines	grid convergence as in S12, `step2` lines 18–20 (sto-3g 0.0010 / 6-31G* 0.0081 / aug-cc-pVDZ 0.0067). *Defined in main.md's concordance but never tagged in the body.*
S14	reference	Lazić, Atodiresei, Alaei, Caciuc, Blügel, Brako, *Comput. Phys. Commun.* **181**, 371–379 (2010), DOI 10.1016/j.cpc.2009.09.016 (volume and pages via Crossref DOI reverse lookup), open preprint arXiv:0810.2273v1; quotation and adjudication in `RESEARCH-20260929-lit.md` §10. Same work as bibkey `lazic2010junolo`, cited above in sections 1 and 4
S15	measurement	retrieval-instrument recall measured in `scratch/oa_stab.js` → `openalex_stability.json` (near-exact title 4/4 at rank 0; the two legitimate topical queries 0/4 each), with the analogous Crossref conclusion in `scratch/xref_calib2.js` (4/4 at rank 0) and `xref_r3.js` (topical 1/3). Includes one self-correction: HTTP 429 throttling was at first mis-scored as "not retrieved"; the conclusion only held after retry with backoff
S16	internal note	`RESEARCH-20260929-lit.md` §8.1 (a direct institutional-network IP reaches the search interface; the `Sign In` prompt is still present, and after the nine queries are submitted the URL does not move and `total=null`)

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